

Sample End Semester Question Paper - 2025

Subject: Digital Signal Processing | Code: EC401 | Semester: 4

Branch: ECE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. What is a window in FIR design?
2. Define linear convolution.
3. What is quantisation?
4. Name two types of digital filters.
5. What is the DTFT?
6. State the sampling theorem.
7. What is aliasing?
8. Define the discrete Fourier transform (DFT).
9. Name one advantage of FFT.
10. What is bilinear transformation?

PART B - Long questions (8 x 10 = 80 Marks)

1. Explain linear and circular convolution with examples.
2. Explain IIR filter structures: direct form and cascade form.
3. Explain effects of finite word length and quantisation.
4. Explain DFT properties and their applications.
5. Explain the sampling theorem and aliasing with examples.
6. Derive the radix-2 DIT FFT algorithm with a flow graph.
7. Explain bilinear transformation for IIR filter design with an example.
8. Design a FIR filter using the window method with an example.